

Cell C5: $d + 2$ lines in $\mathbb{R}^d$

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Official statement (Cell 5). “The case $N = d + 2$ in every dimension, with the same conjectured optimum: the d coordinate axes, two of them repeated. Prove that for every $d \geq 2$, any $d + 2$ lines in $\mathbb{R}^d$ satisfy $S \leq \left(\binom{d+2}{2} - 2\right) \cdot \pi/2$.”

Abstract

We prove that for every $d \geq 1$ (the statement asked for $d \geq 2$), any $d + 2$ lines through the origin of $\mathbb{R}^d$ satisfy $S \leq \left(\binom{d+2}{2} - 2\right) \frac{\pi}{2}$. The proof is unconditional. It has two steps. The first uses the orthogonality-saturation proposition of the C4 notes (`p1_codex/p1_c4.tex`) to reduce an extremal configuration to orthogonal blocks. Each block is an isolated line, a coincident pair, three coincident lines, or a *cycle*. The second step proves a new *cycle lemma* by a monotone potential along a spherical-trigonometric recursion. As by-products, the same argument gives a two-line proof of the $d + 1$ case (C3) and a second proof of the $J = C_6$ case of C4 (six lines in $\mathbb{R}^4$), which the C4 notes settle by their own computation. Every inequality used was checked numerically (Section 5).

1 Setting

Lines are represented by unit vectors $x_1, \dots, x_N \in \mathbb{R}^d$; put $g_{ij} = \langle x_i, x_j \rangle$ and

$$A(x) = \sum_{i < j} \arcsin |g_{ij}|, \quad S(x) = \sum_{i < j} \arccos |g_{ij}| = \binom{N}{2} \frac{\pi}{2} - A(x).$$

Thus $S \leq \left(\binom{d+2}{2} - 2\right) \frac{\pi}{2}$ for $N = d + 2$ is equivalent to $A \geq \pi$. Both quantities depend only on the lines. Two graphs on $\{1, \dots, N\}$ are used: the *orthogonality graph* H ($ij \in H$ iff $g_{ij} = 0$) and its complement J ($ij \in J$ iff $g_{ij} \neq 0$). Only J -edges contribute to A .

2 Structure of an extremal configuration

Proposition 1 (Orthogonality saturation; from the C4 notes, reproved here). *There is a minimiser of A (over N lines in $\mathbb{R}^d$) such that for every i , either $x_i \perp x_j$ for all $j \neq i$, or the vectors $\{x_j : g_{ij} = 0\}$ span a subspace of dimension at least $d - 1$.*

Proof. The configuration space is compact and A is continuous, so the set of minimisers is non-empty. Choose a minimiser with the largest number of orthogonal pairs. Suppose that for some i the space $W = \text{span}\{x_j : g_{ij} = 0\}$ has $\dim W \leq d - 2$ and some $g_{ij} \neq 0$. Since $x_i \perp W$ and $\dim W^\perp \geq 2$, there is a unit $u \in W^\perp$, $u \perp x_i$. Put $x_i(t) = \cos t x_i + \sin t u$ and move only x_i . Pairs (i, j) with $x_j \in W$ stay orthogonal, and pairs not involving i are unchanged. For each j with

$g_{ij} \neq 0$ we have $\langle x_i(t), x_j \rangle = r_j \cos(t - \phi_j)$ with $r_j > 0$, so it vanishes somewhere on the circle but not at $t = 0$. Let $t_- < 0 < t_+$ be the nearest zeros of any of these functions. On (t_-, t_+) each such term $\arccos |r \cos(t - \phi)|$ is convex. Indeed, where $r \cos s > 0$ and $r < 1$,

$$\frac{d^2}{ds^2} \arccos(r \cos s) = \frac{r(1 - r^2) \cos s}{(1 - r^2 \cos^2 s)^{3/2}} \geq 0,$$

and for $r = 1$ the term is $|s|$ near $s = 0$, which is convex. All other terms are constant. Hence $S(x(t))$ is convex on $[t_-, t_+]$, so $S(x(t_+)) \geq S(x(0))$ or $S(x(t_-)) \geq S(x(0))$. That endpoint is again a minimiser of A , and it has strictly more orthogonal pairs. This contradicts the choice of minimiser. $\square$

Lemma 2 (Paths). *Let $y_1, \dots, y_k$ ($k \geq 2$) be unit vectors with $\langle y_i, y_{i+1} \rangle \neq 0$ and $\langle y_i, y_j \rangle = 0$ for $|i - j| \geq 2$. Then $y_1, \dots, y_{k-1}$ are linearly independent.*

Proof. Otherwise let $j \leq k - 1$ be minimal with $y_j \in \text{span}(y_1, \dots, y_{j-1})$ (so $j \geq 2$). Since $y_{j+1} \perp y_1, \dots, y_{j-1}$, it follows that $y_{j+1} \perp y_j$, a contradiction. $\square$

Lemma 3 (Block structure). *Let x be a minimiser as in Proposition 1, and let $N = d + k$.*

1. *Every vertex of J has degree at most k .*
2. *Distinct connected components C, C' of J span mutually orthogonal subspaces $V_C \perp V_{C'}$. Hence $A = \sum_C A_C$, where A_C is the sum over the J -edges of C , and $\sum_C r_C \leq d$, where $r_C = \dim V_C$.*
3. *If a vertex of a component C (with m_C vertices) has J -degree $\delta \geq 1$, then $r_C \leq m_C - \delta$.*

Proof. (1) A vertex of positive J -degree has H -neighbours spanning at least $d - 1$ dimensions, so it has at least $d - 1$ H -neighbours and at most $(N - 1) - (d - 1) = k$ J -neighbours. (2) All pairs across components are H -edges. (3) The H -neighbours of x_i are the vertices of the other components, which span $\bigoplus_{C' \neq C} V_{C'}$ of dimension $\leq d - r_C$, together with the $m_C - 1 - \delta$ non-neighbours of i inside C . By Proposition 1, $d - 1 \leq (d - r_C) + (m_C - 1 - \delta)$. $\square$

3 The cycle lemma

Lemma 4 (Cycle lemma). *Let $m \geq 4$ and let $x_1, \dots, x_m$ be unit vectors (indices mod m) with $\langle x_i, x_j \rangle = 0$ unless $j \in \{i - 1, i, i + 1\}$, $\langle x_i, x_{i+1} \rangle \neq 0$ for all i , and $\dim \text{span}(x_1, \dots, x_m) \leq m - 2$. Then*

$$\sum_{i=1}^m \arcsin |\langle x_i, x_{i+1} \rangle| \geq \pi.$$

For $m = 4$ equality always holds: the four lines are two orthonormal bases of one plane. For $m = 5$ this is the C_5 computation in the C4 notes. For $m = 6$ it is the case $J = C_6$; the C4 notes contain their own complete computation of it, and the cycle lemma gives a second proof.

3.1 Reformulation through a chain basis

By Lemma 2 applied to the path $x_1, \dots, x_{m-1}$, the vectors $b_k := x_k$ ($1 \leq k \leq n := m-2$) are linearly independent. Hence $V := \text{span}(x_1, \dots, x_m) = \text{span}(b_1, \dots, b_n)$ has dimension exactly n . Their Gram matrix is tridiagonal with non-zero off-diagonal entries. We call such a family a *chain basis*. Since $x_{m-1} \in V$ is orthogonal to $b_1, \dots, b_{n-1}$, $x_{m-1} = \pm w_n$, where w_n is the unit normal of $\text{span}(b_1, \dots, b_{n-1})$ in V . Likewise $x_m = \pm u_n$, where u_n is the unit normal of $\text{span}(b_2, \dots, b_n)$ in V . (For $n = 1$ we use the convention $u_1 = w_1 = b_1$: the orthogonal complement of $\text{span}(\emptyset) = \{0\}$ in $\mathbb{R}b_1$ is $\mathbb{R}b_1$. This is what the case $n = 1$ of Lemma 5 uses.) For a chain basis define $X_n, Y_n, \eta_n \in [0, \pi/2]$ and s_n by

$$\cos X_n = |\langle b_1, u_n \rangle|, \quad \cos Y_n = |\langle b_n, w_n \rangle|, \quad \sin \eta_n = |\langle u_n, w_n \rangle|, \quad s_n = \sum_{k=1}^{n-1} a_k, \quad a_k = \arcsin |\langle b_k, b_{k+1} \rangle|.$$

Here $\cos X_n, \cos Y_n > 0$ by independence. The m edges of the cycle contribute s_n , $\arcsin \cos Y_n = \frac{\pi}{2} - Y_n$ (edge $x_{m-2}x_{m-1}$), η_n (edge $x_{m-1}x_m$) and $\frac{\pi}{2} - X_n$ (edge x_mx_1). Thus

$$\sum_i \arcsin |\langle x_i, x_{i+1} \rangle| - \pi = \Phi_n := s_n - X_n - Y_n + \eta_n, \quad (1)$$

and the cycle lemma follows once we show $\Phi_n \geq 0$ for all chain bases with $n \geq 2$.

3.2 The recursion

Lemma 5. *Let $b_1, \dots, b_{n+1}$ be a chain basis ($n \geq 1$) and let $a = a_n$. Then*

$$\sin a = \cos Y_n \sin Y_{n+1}, \quad (R1)$$

$$\tan \eta_{n+1} = \tan Y_{n+1} \sin \eta_n, \quad (R2)$$

$$\cos X_{n+1} = \cos X_n \cos \eta_{n+1}. \quad (R3)$$

The initial values are $X_1 = Y_1 = 0$ and $\eta_1 = \pi/2$. Consequently $X_2 = Y_2 = \eta_2 = a_1$ and $\Phi_2 = 0$. For $n \geq 2$ we have $\eta_n \leq Y_n$ and $\eta_n \leq X_n$.

Proof. Let $E = \text{span}(b_1, \dots, b_n)$ and let e be the unit normal of E in $E \oplus \mathbb{R}b_{n+1}$, so $w_{n+1} = \pm e$. Since $b_{n+1} \perp b_1, \dots, b_{n-1}$, its projection to E is parallel to w_n . After changing signs of w_n, e (this only changes the sign of $c = \langle u_n, w_n \rangle$ below, and only $|c|$ is used) we can write $b_{n+1} = p w_n + q e$ with $p, q \geq 0$ and $p^2 + q^2 = 1$, and $q > 0$ by independence. Then $\cos Y_{n+1} = q$ and $p = \sin Y_{n+1}$. Moreover $\langle b_n, b_{n+1} \rangle = p \langle b_n, w_n \rangle$, which gives (R1). The orthogonal complement of $\text{span}(b_2, \dots, b_n)$ is $\text{span}(u_n, e)$, so $u_{n+1} = \alpha u_n + \beta e$ with $\alpha^2 + \beta^2 = 1$ and $0 = \langle u_{n+1}, b_{n+1} \rangle = \alpha p c + \beta q$, where $c = \langle u_n, w_n \rangle$ and $|c| = \sin \eta_n$. Hence $(\alpha, \beta) = (q, -pc)/\rho$ with $\rho = \sqrt{q^2 + p^2 c^2} > 0$. Then $\sin \eta_{n+1} = |\beta|$ and $\cos \eta_{n+1} = |\alpha|$, which gives (R2). Also $\cos X_{n+1} = |\langle b_1, u_{n+1} \rangle| = |\alpha| \cos X_n$, because $b_1 \perp e$. This gives (R3). The initial values follow from $u_1 = w_1 = b_1$. Finally, (R2) gives $\tan \eta_{n+1} \leq \tan Y_{n+1}$ and (R3) gives $\cos X_{n+1} \leq \cos \eta_{n+1}$. $\square$

3.3 Monotonicity of the potential

Lemma 6 (Two right triangles). *For $\zeta \in [0, \pi/2]$ and $Y' \in [0, \pi/2)$ let $\zeta^* \in [0, \pi/2)$ be defined by $\tan \zeta^* = \tan Y' \sin \zeta$. Then*

$$K(\zeta, Y') := \arcsin(\sin Y' \cos \zeta) + \zeta - Y' - \arccos(\cos \zeta \cos \zeta^*) + \zeta^* \geq 0.$$

Proof. If $\zeta \in \{0, \pi/2\}$ or $Y' = 0$, a direct substitution gives $K = 0$. So let $\zeta, Y' \in (0, \pi/2)$, and put $s = \sin \zeta \in (0, 1)$, $c = \cos \zeta = \sqrt{1 - s^2}$ and $f(x) = \arccos(c \cos x) - x$ on $[0, \pi/2]$. With $x_1 = \zeta^*$ and $x_2 = \pi/2 - Y'$ we have $\arcsin(\sin Y' \cos \zeta) = \frac{\pi}{2} - \arccos(c \cos x_2)$. Hence

$$K = \zeta - f(x_1) - f(x_2), \quad \tan x_1 \tan x_2 = s.$$

Write $\tan x_1 = \sqrt{s} e^v$, $\tan x_2 = \sqrt{s} e^{-v}$ ($v \in \mathbb{R}$) and $F(v) = h(v) + h(-v)$ with $h(v) = f(\arctan(\sqrt{s} e^v))$. F is even and $F(v) \rightarrow f(\pi/2) + f(0) = 0 + \zeta$ as $v \rightarrow \infty$. It therefore suffices to show $F'(v) \geq 0$ for $v > 0$. With $T = \tan x$ one computes $dx/dv = T/(1 + T^2)$ and $f'(x) = cT/\sqrt{T^2 + s^2} - 1$, so

$$h'(v) = q(T) := \frac{T}{1 + T^2} \left(\frac{cT}{\sqrt{T^2 + s^2}} - 1 \right), \quad F'(v) = q(t) - q(s/t), \quad t = \sqrt{s} e^v > \sqrt{s}.$$

With $\mathcal{A} = \sqrt{1 + t^2}$ and $\mathcal{B} = \sqrt{t^2 + s^2}$ we have $q(s/t) = \frac{st}{\mathcal{B}^2} \left(\frac{c}{\mathcal{A}} - 1 \right)$ and

$$q(t) - q(s/t) = \frac{t}{\mathcal{A}^2 \mathcal{B}^2} \left[s\mathcal{A}^2 - \mathcal{B}^2 - c(s\mathcal{A} - t\mathcal{B}) \right] = \frac{t(t^2 - s)}{\mathcal{A}^2 \mathcal{B}^2} \left[\frac{c(t^2 + s)}{t\mathcal{B} + s\mathcal{A}} - (1 - s) \right],$$

using $s\mathcal{A}^2 - \mathcal{B}^2 = -(1 - s)(t^2 - s)$ and $t\mathcal{B} - s\mathcal{A} = (t^4 - s^2)/(t\mathcal{B} + s\mathcal{A})$. Since $t^2 > s$, it remains to show that $\sqrt{1 + s}(t^2 + s) \geq \sqrt{1 - s}(t\mathcal{B} + s\mathcal{A})$. First, $\sqrt{(1 + s)/(1 - s)} \geq 1 + s$, because this is equivalent to $1 \geq 1 - s^2$. Next, $t\mathcal{B} = t^2 + \frac{ts^2}{\mathcal{B} + t} \leq t^2 + \frac{s^2}{2}$ and $s\mathcal{A} = s + \frac{st^2}{\mathcal{A} + 1} \leq s + \frac{st^2}{2}$. Therefore $t\mathcal{B} + s\mathcal{A} \leq (t^2 + s)(1 + \frac{s}{2}) \leq (1 + s)(t^2 + s)$. $\square$

Proposition 7. *For every chain basis and every $n \geq 2$, $\Phi_{n+1} \geq \Phi_n$. Hence $\Phi_n \geq \Phi_2 = 0$, and the cycle lemma holds.*

Proof. Write X, Y, η for step n , primes for step $n + 1$, and $a = a_n$. Then $\Phi_{n+1} - \Phi_n = a - (X' - X) - (Y' - Y) + (\eta' - \eta)$.

(i) *Eliminating X .* By (R3), $X' - X = \varphi(X)$ with $\varphi(\xi) = \arccos(\cos \xi \cos \eta') - \xi$. This function is non-increasing on $[0, \pi/2]$, because $\varphi'(\xi) = \sin \xi \cos \eta' / \sqrt{1 - \cos^2 \xi \cos^2 \eta'} - 1 \leq 0$ (the radicand exceeds $\sin^2 \xi \cos^2 \eta'$ by $\sin^2 \eta' \geq 0$). Since $X \geq \eta$ (Lemma 5), $X' - X \leq \arccos(\cos \eta \cos \eta') - \eta$. Hence

$$\Phi_{n+1} - \Phi_n \geq \sigma(Y, Y') - [\arccos(\cos \eta \cos \eta') - \eta'], \quad \sigma(Y, Y') := \arcsin(\sin Y' \cos Y) + Y - Y',$$

using $a = \arcsin(\sin Y' \cos Y)$ from (R1).

(ii) *Replacing Y by η .* We have $\partial_Y \sigma = 1 - \sin Y' \sin Y / \sqrt{1 - \sin^2 Y' \cos^2 Y} \geq 0$, since $1 - \sin^2 Y' \cos^2 Y - \sin^2 Y' \sin^2 Y = \cos^2 Y' \geq 0$. Because $\eta \leq Y$, it follows that $\sigma(Y, Y') \geq \sigma(\eta, Y')$.

(iii) By (R2), η' is exactly ζ^* for $\zeta = \eta$. Therefore $\Phi_{n+1} - \Phi_n \geq K(\eta, Y') \geq 0$ by Lemma 6. Combined with (1), this proves Lemma 4. $\square$

4 Main theorem

Theorem 8. *For every $d \geq 1$, any $d + 2$ unit vectors in $\mathbb{R}^d$ satisfy $\sum_{i < j} \arcsin |\langle x_i, x_j \rangle| \geq \pi$. Equivalently, any $d + 2$ lines in $\mathbb{R}^d$ have $S \leq \left(\binom{d+2}{2} - 2 \right) \frac{\pi}{2}$.*

Proof. It suffices to bound A at the minimiser of Proposition 1. By Lemma 3 with $k = 2$, every component C of J is an isolated vertex, a path or a cycle. Let $\nu_C = m_C - r_C$. We claim that $A_C \geq \nu_C \pi/2$ in every case.

- Isolated vertex: $\nu_C = 0$.

- Path on two vertices: $r_C \leq 1$ by Lemma 3(3), so the two lines coincide. Then $\nu_C = 1$ and $A_C = \pi/2$.
- Path on $k \geq 3$ vertices: Lemma 2 gives $r_C \geq k - 1$, while an interior vertex gives $r_C \leq k - 2$. This case is impossible.
- Triangle: $r_C \leq 1$, so the three lines coincide. Then $\nu_C = 2$ and $A_C = 3\pi/2$.
- Cycle of length $m \geq 4$: $r_C \leq m - 2$ by Lemma 3(3) and $r_C \geq m - 2$ by Lemma 2, so $\nu_C = 2$. Lemma 4 gives $A_C \geq \pi$.

Finally $\sum_C \nu_C = N - \sum_C r_C \geq (d + 2) - d = 2$, so $A = \sum_C A_C \geq \pi$. $\square$

Corollary 9 (C4). *Five lines in $\mathbb{R}^3$ and six lines in $\mathbb{R}^4$ satisfy $A \geq \pi$.*

Corollary 10 (C3, without the prefix lemma). *Any $d + 1$ unit vectors in $\mathbb{R}^d$ satisfy $\sum \arcsin |g_{ij}| \geq \pi/2$.*

Proof. Lemma 3 with $k = 1$ shows that every component of J is an isolated vertex ($\nu = 0$) or a single edge with $r \leq 1$, i.e. a coincident pair ($\nu = 1$, $A_C = \pi/2$). Since $\sum \nu_C \geq 1$, the bound follows. $\square$

Remark 11 (Equality and extensions). The value π is attained, for example by the following configurations:

- two disjoint coincident pairs completed by an orthonormal basis;
- two orthonormal bases of a plane (the 4-cycle), completed by an orthonormal basis of the orthogonal complement;
- for $d \geq 4$, two “planar $0^\circ, 45^\circ, 90^\circ$ triples” in orthogonal planes, completed by an orthonormal basis of the rest.

The last example contains a J -path on three vertices. This does not contradict the exclusion of such paths in the proof of Theorem 8, because that minimiser is not saturated: that exclusion applies only to the minimiser chosen in Proposition 1. In our numerical searches ($d = 2, \dots, 5$), every minimiser found was of these block types. This is an observation, not a classification.

For $N = d + k$, the same saturation step reduces the Fejes Tóth conjecture for $N \leq 2d$ to proving $A_C \geq \nu_C \pi/2$ for components of J with maximum degree $\leq k$ and $r_C \leq m_C - \deg$. For $k \geq 3$ these components are no longer just paths and cycles. The potential Φ of Proposition 7 was found numerically and may extend.

5 Numerical verification

All scripts are in `p1_c5/`.

- `direct_min.py`: direct minimisation of A for $d + 2$ lines, $d = 2, \dots, 5$, with 120–200 random starts each. The minimum is π to 10^{-8} , no value below π was found, and the minimisers have the block structure above.
- `cycle.py`, `concavity.py`: an explicit parametrisation of the cycle family ($x_1 = e_1$, $x_{k+1} = \sin t_k e_k + \cos t_k e_{k+1}$). It confirms the orthogonality pattern and gives $\min(A - \pi) \geq -10^{-8}$ for $m = 4, \dots, 12$ (random sampling and local optimisation). The minimum is approached only at the degenerate boundary.

- `recursion.py`: (R1)–(R3) agree with the explicit parametrisation, and $\Phi_{n+1} \geq \Phi_n$ holds along 10^5 random chains.
- `verify_all.py` (10^6 samples each, float64) and `verify_hp.py` (50-digit mpmath): these check the convexity in Proposition 1, the invariants $\eta \leq Y$ and $\eta \leq X$, monotonicity of φ and σ , the inequality $q(t) \geq q(s/t)$, the final algebraic inequality, $K \geq 0$, the full step inequality, and the identity (1) ($|A - \pi - \Phi_n| \leq 2 \cdot 10^{-33}$ in high precision). The float64 violations of order 10^{-9} were arccos cancellation. They disappear at 50 digits.